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ज्ञानम् सकलजनहिताय

An Autonomous Institute Affiliated with
Savitribai Phule Pune University

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A Project Phase-II Report

On

Sustainable Hydroponic Farming

Intelligent Automation and Monitoring

Submitted in the partial fulfillment of the requirements of

Bachelor of Engineering

in

Electronics and Telecommunication

By

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Under the guidance of

Dr. R. R. Itkarkar

Project Sponsored by: **Apollo Automations**

Department of Electronics and Telecommunication Engineering

All India Shri Shivaji Memorial Society's College of Engineering, Pune-01

Affiliated to SPPU Pune

A.Y 2025-26



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CERTIFICATE

This is to certify that the Project titled “**Sustainable Hydroponic Farming Intelligent Automation and Monitoring.**”

Project Sponsored by: Apollo Automations

Submitted by,

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is a Bonafide record of the Project Phase-II work carried out by them towards the partial fulfillment of the requirements of Savitribai Phule Pune University, for the award of Bachelor of Electronics and Telecommunication Engineering under my supervision and guidance.

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Dr.S.H.Lavate

Project coordinator

Dr. S B Dhonde

HOD- E & TC

Date:

Place: **Pune**

This report has been examined as per Savitribai Phule Pune University requirements at **AISSMS COE, Pune-01** on _____

Signature:

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SPONSERSHIP LETTER



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To
The Principal
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No. 1, Kennedy Road, Near RTO Office, Sangamvadi, Shivajinagar,
Pune, Maharashtra 411001

Subject: Sponsorship letter for Final Year Students of AISSMS COE, Pune.

Respected Sir/Madam,

We at Appolo AI Technologies, Pune, are keen to extend our support towards the academic and professional development of final-year students in the Electronics and Telecommunication Department of AISSMS COE, Pune, for their major project.

We intend to sponsor selected students involved in the Hydroponic Automation Project for the academic year 2025-26, providing financial assistance, technical mentorship, and resources essential for successful project execution.

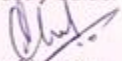
We assure complete cooperation and support for the completion of the project. We look forward to collaborating with AISSMS COE, Pune, to strengthen the academia and industry interaction for student benefit.

Project Title: Sustainable Hydroponic Farming, through Intelligent Monitoring and Automation
Institute Guide: Dr. R.R.Itkarkar

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Thanking you,
CEO, Apollo Automations
Amol Chawathe

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We would like to express our sincere gratitude and appreciation to all those who have contributed to the successful completion of this engineering project.

First and foremost, we extend our heartfelt thanks to our project mentor, Dr. R. R. Itkarkar, for their invaluable guidance, continuous support, and expertise throughout the project. Their insightful feedback and constructive suggestions were instrumental in shaping our work. We also express our gratitude towards our project coordinator Dr. S.H. Lavate sir for his kind support and co-operation. We are grateful to Principal Dr. D. S. Bormane and Dr. S.B. Dhonde Head of E &TC Engineering Department and all the teaching and non-teaching staff members of E &TC Engineering Department for their direct or indirect help in the completion of the project.

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PROLOGUE

All India Shri Shivaji Memorial Society, established as early as in 1917, is a premier educational institution in Pune. Ours is a result-oriented Society dedicated to the noble cause of Military, General, Technical & Management Education in India by Shri Chhatrapati Shahu Maharaj of Kolhapur also known as "Rajarshi Shahu". He was the first Maharaja of the princely state of Kolhapur and a great social reformer.

He was an invaluable gem in the history of Maharashtra he worked tirelessly for the cause of the lower caste subjects in his state. Primary education to all regardless of caste and creed was one of his most significant priorities. In our Institute we follow the noble practices established by our great leader and as we know Leader decide the destiny we are blessed to have a leadership and a rich cultural heritage which is totally focused on the noble cause of Quality Education. It functions beyond race, caste, creed, religion & political spirit. The Society Management is very pragmatic & progressive.

On Nov 10, 1917, Rajarshi Chhatrapati Shahu Maharaj announced in Delhi, his proposal to establish a Memorial of Chhatrapati Shivaji Maharaj in the City of Pune. In the proposal, he declared his intention of creating a central hall with Chhatrapati Shivaji Maharaj's life size statue and a hostel to accommodate a hundred Maratha students. The institute was to act as a rallying centre for Maratha activities all over India, giving a concrete shape and impetus to the diverse endeavours for the advancement of the Maratha community.

On Dec 27, 1917, Rajarshi Chhatrapati Shahu Maharaj, on a pressing demand from Khaserao Jadhav,

presided over the 11th session of the Maratha Educational Conference at Khamgoan. Rajarshi Chhatrapati Shahu Maharaj had convened a meeting of the committee of the All India Shivaji Memorial Society on 25 May 1922. Chhatrapati Rajaram Maharaj was elected President of the All India Shivaji Memorial Society and took on.

Core Values :

- Leadership and cultural heritage
- Honesty and Integrity
- Freedom of thought and Expression
- Excellence
- Accountability and Transparency
- Encouragement
- Social Responsibility

Honesty & Integrity :

The foreword of our society is "Satyala Maran Nahi" which means "Truth is eternal"



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COLLEGE VISION AND MISSION

Vision

Service to society through quality education.

Mission :

- Generation of national wealth through education and research
- Imparting quality technical education at the cost affordable to all strata of the society
- Enhancing the quality of life through sustainable development
- Carrying out high quality intellectual work
- Achieving the distinction of highest preferred engineering college in the eyes of the stake holders

Goal :

- To inculcate learning habits
- To create an environment to make the students creative and innovative
- To promote project based learning
- To strengthen industry – institute interaction
- To ensure continuous improvement in quality
- To develop entrepreneurship skills
- To nurture the spirit of team work
- To catalyse all – round development of students
- To develop technologies for sustainable development

DEPARTMENT VISION AND MISSION

Vision :

Society Growth and Welfare Through Competent Electronics and Communication Engineering Graduates.

Mission :

- To impart quality education in the field of E & TC engineering to solve societal and industrial problems with focus on trans-disciplinary approach
- To provide stimulating learning environment with modern tools & technologies.
- To produce dynamic graduates with ethics and moral values.
- To facilitate E & TC graduates with sight of innovation



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Course Outcomes :

Course outcomes -Project Stage-1

CO1: Identify project for society and industry need by applying engineering knowledge gained throughout the E and TC Engineering program

CO2: Investigate identified complex engineering problem using appropriate research methods and techniques

CO3: Test the solution of identified engineering problem with appropriate simulation tool.

CO4: Work in team and effective budget Planning to meet the project requirement

CO5 : Effectively communicate the project progress through presentations and technical report.

CO6: Develop self-learning skills and follow the ethical code of conduct for project.

Course outcomes -Project Stage-2

CO1: Develop solutions to the real world problems using modern engineering tools and technologies.

CO2: Demonstrate practical skills and knowledge in testing and debugging for both hardware and software based projects.

CO3: Work in team to demonstrate the project by using visual aids and visualization techniques.

CO4: Effectively communicate project work through publications, competitions, presentations and technical report.

CO5: Showcase the project management and self-learning skills for lifelong learning.

CO6: Adherence to ethical code of conduct for project execution.

Academic Year : 2025- 26

Name of the Programme: Electronics & Telecommunication Engineering.

Program Outcomes :

1. A graduate student will apply knowledge of mathematics and engineering fundamentals to design electronic circuits and systems such as logical circuits, analog circuits, electrical machines, control and communication systems, digital systems etc. and test the results.
2. A graduate student will be able to identify, formulate, solve electronics engineering problems.
3. A graduate student will demonstrate the ability to design, implement and evaluate a system, process, component and program to meet the specified needs with appropriate considerations for public, health and safety, society and environment.
4. A graduate student will investigate, formulate, analyze and provide appropriate solution to simple and complex engineering problem.
5. A graduate student will provide the solutions by using the Modern electronic and IT Engineering Tools and Technologies for practicing electronic Engineering problems
6. A graduate student will demonstrate the ability to learn the impact of industries on society by visiting different industries and understand the importance of industrial products for analog and digital circuits and systems.
7. A graduate student will understand the importance of environmental issues and will design sustainable systems.
8. A graduate student will understand the professional and ethical responsibilities to meet the socio- economic challenges.
9. A graduate student will function effectively as an individual, member or leader of a team in multidisciplinary setting.
10. A graduate student will be able to communicate effectively at different technical and administrative levels.
11. A graduate student will demonstrate knowledge and understanding of engineering and management principles to manage projects.
12. A graduate student will recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcome (PSO's) of Department

1. Systems Development: Design and develop prototype or system using hardware and software skills for multi-disciplinary applications.
2. Adaptability: Demonstrate an ability to adapt emerging technologies in the field of Artificial Intelligence machine learning, Internet of things and Robotics to solve the real life problems.
3. Skill Development: Develop technical and management skills through student clubs and industry interactions.

List of Program Educational Objectives (PEO's)

1. To build strong fundamental knowledge among graduates required to pursue their higher education and continue professional development.
2. To enable graduates to identify, analyze and solve Electronics Engineering problems by applying basic principles and modern techniques.
3. To enable graduates to innovate, design and develop hardware & software components and groom their ability to succeed in multidisciplinary & diverse field.
4. To inculcate in graduates professional attitude, effective communicational skills, team work skills for becoming a responsible, cultured human being.

ABSTRACT

The increasing demand for food production, coupled with limited agricultural land and water scarcity, has created a need for sustainable farming techniques. Hydroponic farming, a soilless cultivation method in which plants are grown using nutrient-rich water solutions, has emerged as an efficient alternative to conventional agriculture. It offers significant advantages such as reduced water consumption, improved crop yield, and controlled cultivation conditions.

This project presents the design and implementation of an IoT-based intelligent hydroponic farming system using the ESP32 microcontroller. The system integrates various sensors, including a pH sensor, TDS sensor, DHT22 temperature and humidity sensor, and water level sensor, to continuously monitor critical parameters required for plant growth. The collected data is processed by the ESP32 and compared with predefined threshold values to maintain optimal growing conditions.

The system is integrated with the Blynk IoT platform, enabling real-time monitoring through a mobile application. Users can view live sensor readings and receive notifications and email alerts when the water level falls below the required threshold. The automation and monitoring features improve system reliability while reducing manual intervention.

Experimental testing demonstrated successful monitoring of pH, TDS, temperature, humidity, and water level parameters. The system also generated real-time notifications and email alerts when the water level dropped below the predefined threshold. The developed system provides a practical and cost-effective solution for smart and sustainable agriculture.

Abbreviations

Abbreviation	Full Form
IoT	Internet of Things
ESP32	Espressif 32-bit Microcontroller
TDS	Total Dissolved Solids
OLED	Organic Light Emitting Diode
DHT	Digital Humidity and Temperature Sensor
pH	Potential of Hydrogen
Wi-Fi	Wireless Fidelity
LCD	Liquid Crystal Display
AI	Artificial Intelligence
ML	Machine Learning
IIoT	Industrial Internet of Things
GPIO	General Purpose Input Output
I2C	Inter-Integrated Circuit
UART	Universal Asynchronous Receiver Transmitter
SPI	Serial Peripheral Interface
RH	Relative Humidity
DC	Direct Current
NFT	Nutrient Film Technique
DWC	Deep Water Culture
EC	Electrical Conductivity

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Agriculture has always been one of the most important sectors for human survival and economic development. A large portion of the global population depends on agriculture for food production and livelihood. However, traditional farming methods are facing several challenges due to rapid urbanization, climate change, water scarcity, decreasing availability of fertile land, and increasing population growth. These challenges have created a strong need for modern agricultural techniques that can improve productivity while efficiently utilizing available resources. Conventional soil-based farming requires large amounts of water, fertilizers, and land for cultivation. In many regions, irregular rainfall patterns and depletion of groundwater resources have made agriculture increasingly difficult. Additionally, soil degradation, pest attacks, and dependency on climatic conditions often reduce crop quality and yield. As the demand for food continues to increase, sustainable and resource-efficient farming solutions have become essential.

Hydroponic farming has emerged as an effective alternative to traditional agriculture. Hydroponics is a soilless cultivation technique in which plants are grown using nutrient-rich water solutions instead of soil. In this method, essential nutrients are supplied directly to the plant roots in controlled quantities, allowing faster growth and better nutrient absorption. Since the system recirculates water, hydroponics consumes significantly less water compared to conventional farming methods. It is highly suitable for urban farming, indoor cultivation, greenhouse environments, and regions affected by water scarcity.

Although hydroponic farming offers several benefits, maintaining optimal conditions manually can be difficult. Parameters such as pH level, nutrient concentration, temperature, humidity, and water level must be continuously monitored and controlled to ensure healthy plant growth. Manual monitoring is time-consuming, less accurate, and prone to human error, especially in larger systems. The advancement of technologies such as the Internet of Things (IoT), embedded systems, and wireless communication has introduced new possibilities in smart agriculture. IoT-based systems enable real-time monitoring, automation, and remote access to agricultural operations. By integrating IoT with hydroponic farming, it becomes possible to automate environmental monitoring and maintain optimal conditions with minimal human intervention. To address these challenges, the proposed project has been developed. The system utilizes an ESP32 microcontroller integrated with various sensors to continuously monitor critical hydroponic parameters. The collected data is transmitted to the Blynk IoT platform through Wi-Fi, enabling real-time monitoring through a mobile application. The system also generates notification and email alerts whenever the water level falls below the predefined threshold, thereby improving reliability and reducing manual intervention.

1.2 NEED OF HYDROPONICS

The population increase and urbanization have caused an increased demand for food production while simultaneously decreasing fertile land available for agriculture. Conventional methods of farming require high amounts of water and rely on the quality of soil and climate. Water shortage and degraded soil quality are major obstacles in conventional farming in most areas today. This problem has necessitated new approaches to farming to make use of minimal space yet generate higher outputs efficiently.

Hydroponic farming offers solutions to these problems. By growing crops in water instead of soil, hydroponics becomes less reliant on fertile soils and can even be done indoors, in urban areas, and even in soils that do not meet conventional farming requirements. Hydroponic farming also helps overcome difficulties associated with weed, soil diseases, and pests. One of the main benefits of hydroponic farming is its efficient use of water resources. As opposed to conventional farming, hydroponic farms continuously recirculate water, meaning that water requirements are significantly lower. This feature makes hydroponic farming very beneficial in areas with water shortages. Moreover, hydroponic plants absorb nutrients from the water solution, which leads to better nutrient intake and faster plant growth. Hydroponics not only enables year-round production but also facilitates the control of the environment throughout the growth period of the crop. The pH, nutrient levels, temperature, and humidity can all be better managed, ensuring better quality and efficiency. This makes hydroponics a valuable technology for the future of agriculture.

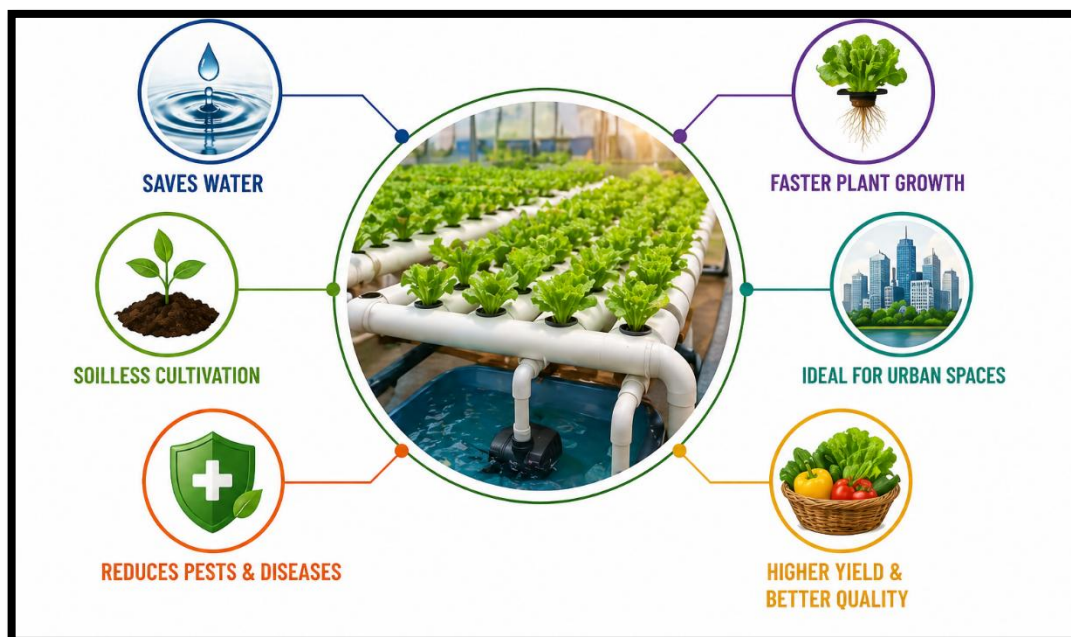


Fig 1.1: Need of Hydroponics

CHAPTER 2

PROBLEM STATEMENT, AIM and OBJECTIVES

2.1 PROBLEM STATEMENT

Manual monitoring and control of hydroponic systems are inefficient, error-prone, and lack real-time decision-making, leading to suboptimal plant growth and resource wastage.

2.2 AIM

To design and develop an IoT-based hydroponic farming system for real-time monitoring and control of plant growth parameters.

2.3 OBJECTIVES

- To implement real-time monitoring of pH, temperature, humidity, and water level.
- To generate real-time notification and email alerts when water level falls below threshold.
- To automate water flow using pumps.
- To develop a system capable of assisting urban farming and sustainable agriculture.
- To provide wireless visualization of system data using Blynk IoT and OLED display.

CHAPTER 3

LITERATURE SURVEY

3.1 LITERATURE SURVEY

Papers

Sr No	Title	Authors	Year	Research Gap	Publication Source
1	A Sensor-Driven Automated Hydroponic System for Optimized Plant Growth in Diverse Environments	Joshitha C, Nitin, Bobba Pandu Ranga, Bandi Harshini Reddy	2025	Lack of real-time automation and environmental control	IEEE IITCEE
2	An IoT Monitoring System Designed for Hydroponics Plant Cultivation	Eric Blancaflor, Jameela Jamena, Kevin Banganay, Raya Rabanal, Karen Fernandez, Stephany Zamora	2022	No efficient remote monitoring system	IEEE ICCBD
3	Automated Hydroponic Vertical Farming using IoT	K. Ezhilarasan, Shilpa A N, Sridhar N	2025	Space limitation and inefficient resource usage	IEEE ICKECS
4	Automating Hydroponics: Enhance the Crop Cultivation and Monitoring Using IoT and ML	G. Brindha, Devadharshini D, Kumaragurubaran T, Dhanush S	2024	Lack of intelligent nutrient optimization	Journal / Conference
5	Development of an IoT-Enabled Real-Time Monitoring and Alert System Model for Hydroponic Farming	Palash Gourshettiwar, KTV Reddy	2025	Absence of real-time alert mechanisms	IEEE ICTMIM
6	Enhancing Hydroponic Farming using IoT Integration and Machine Learning Techniques	Dr. C. Suresh, Selvabhuvaneswari S	2025	Low efficiency due to lack of predictive control	IEEE ICMSCI
7	Enhancing Hydroponic Production: A Review of Plant Growth and Health Monitoring System	Ajay Basil Varghese, Dr. Deepika M P	2024	Lack of integrated monitoring systems	IEEE ICICI

Sr No	Title	Authors	Year	Research Gap	Publication Source
8	Enhancing Hydroponic Systems with IoT: Optimizing pH, Temperature, and Humidity Parameters	Subhra Debdas, Akshat Jain, Anika Mariam, Anurati Maiti, Anushka Borah	2024	Difficulty in maintaining stable environmental conditions	Journal / Conference
9	IoT Integrated Hydroponic System for Precision Agriculture	Senthil Priya R M, Varsha V, Raghavi S, Rajalakshmi A N	2025	Lack of intelligent decision-making systems	Journal / Conference
10	Real Time Monitoring of Hydroponic System using IoT	Vrushali Waghmare, Perna Rakte, Meghana Gadalkar, Sakshi Bhaleghare, Dipti Sakhare, Mahesh Goudar	2023	Inefficient traditional monitoring systems	IEEE
11	Smart Hydroponic System: AI and IoT-Powered Fertilizer and Plant Health Monitoring	P. Suman Prakash, Shaik Afsha Jabeen, Lalith Venkat, Yashwanth Sagar, Mohammed Ismail.	2025	Nutrient imbalance due to manual control	Journal / Conference
12	Smart IoT based Nutrition Alert System for Hydroponic Plant Growth	G. Gokulraj, K. M. Kirthika, R. N Devendra Kumar, Balasubramaniam Vadivel, M. Suresh Kumar, V. Padmacharan	2025	Lack of timely nutrient correction alerts	Journal / Conference
13	Urban Vertical Farming with IoT and Precision Monitoring for a Sustainable Future	Unni Krishnan A, Amit Dutt, Jay Singh, Munugapati Bhavana, Swathi Baswaraju, Baydaa Fathil Mohammed, Akila Venkatraman	2025	Urban food demand and limited land	Journal / Conference
14	Vertical Farming with Smart Technology to Increase Agricultural Productivity	Narumol Chumuang	2025	Low productivity in traditional farming	Journal / Conference

Journals

Sr. No.	Title	Key Contributors (Authors)	Year	Key Contribution
1	Elucidation of Intelligent Classification Framework for Hydroponic Lettuce Deficiency Using Enhanced Optimization Strategy and Ensemble Multi-Dilated Adaptive Networks	Mustufa Haider Abidi, Sanjay Chintakindi, Ateekh Ur Rehman, Muneer Khan Mohammed	2024	Developed deep learning-based system for detecting nutrient deficiencies in hydroponic lettuce
2	Field Study Correlating Nutrient Absorption and Transpiration in Lettuce Hydroponics Using an IoT-Interfaced Solid-State Ion Sensor Array	Yi-Min Wu, Shao-Yuan Liu, Bo-You Shi, Jui-Yu Peng, Zhi-Wei Kao, Hsing-Ying Chung, Chi-Yi Lin, Wei Fang, Lin-Chi Chen	2024	Developed IoT-based ion sensor system to monitor real-time nutrient absorption
3	IIoT Based Multimodal Communication Model for Agriculture and Agro-Industries	Basem Almadani, Saud Mohammad Mostafa	2021	Proposed communication model for integrating multiple agricultural systems using IIoT
4	Improving Hydroponic Systems by Using ARUCO Markers for Leaf Detection: Focus on Tomato Plants	P. Tharun, S. Muthulakshmi, N. Subhashini, P. Vishnuram, R. Kesavan	2025	Introduced image processing technique for plant monitoring using ARUCO markers
5	Internet of Things-Based Smart Precision Farming in Soilless Agriculture: Opportunities and Challenges for Global Food Security	Monica Dutta, Deepali Gupta, Sumegh Tharewal, Deepam Goyal, Jasminder Kaur Sandhu, Manjit Kaur, Ahmad Ali Alzubi, Jazem Mutared Alanazi	2025	Comprehensive study on IoT-based smart farming and its impact on food security
6	Revolutionizing Holy-Basil Cultivation With AI-Enabled Hydroponics System	G. Lakshmi Priya, Chanthini Baskar, Sanket Sandeep Deshmane, C. Adithya, Souranil Das	2023	Developed AI + IoT-based hydroponic system for medicinal plants

3.2 LITERATURE SURVEY SUMMARY

[1] The paper describes an automated hydroponic system that employs sensors and microcontrollers to detect and regulate various factors such as pH, TDS, temperature, humidity, and water level. The system uses an ESP32 design that can be used indoors or outdoors to automate and minimize human interventions in environmental controls. Such systems increase productivity while working under varied conditions. However, the system lacks features that offer advanced analytical capabilities, does not include intelligent approaches like AI, and does not solve the problem of scalability.

[2] The study focuses on developing a hydroponic monitoring system using IoT. It integrates multiple sensors to track parameters such as pH, EC, temperature, humidity, and water levels, with data visualization through the Blynk mobile application. The system enhances accessibility and allows remote monitoring of plant conditions. However, the system mainly emphasizes monitoring rather than automation, lacks a closed-loop control mechanism, and depends heavily on continuous internet connectivity for operation.

[3] The paper introduces a vertical farming model integrated with IoT technologies. It uses sensors like DHT11 and LDR along with a NodeMCU controller to monitor environmental conditions and automate irrigation. The system supports remote monitoring through cloud platforms like ThingSpeak and improves space utilization in farming. However, it uses basic sensors with limited accuracy, lacks detailed nutrient monitoring such as EC and pH precision, and is dependent on continuous power supply for efficient operation.

[4] The research integrates machine learning with IoT to optimize nutrient delivery and environmental control. Using algorithms such as Random Forest and Neural Networks, the system analyzes sensor data and predicts optimal nutrient levels dynamically. This improves crop yield and resource efficiency significantly. However, the system requires large datasets, high computational power, and complex implementation, making it less suitable for cost-effective and small-scale applications.

[5] The paper proposes a cloud-based hydroponic monitoring system. It continuously measures environmental parameters and provides real-time alerts using APIs and cloud services. The system ensures efficient monitoring and improves plant growth conditions. However, it relies heavily on cloud infrastructure, introduces latency due to data transmission, and lacks strong local control mechanisms.

[6] The study combines IoT with machine learning to optimize hydroponic farming processes. It uses sensor data and predictive algorithms to automatically adjust nutrient delivery, lighting, and environmental conditions. The system demonstrates improved crop yield and reduced resource consumption. However, the implementation is complex, requires trained datasets, and increases overall system cost, limiting its practical usability for small-scale farmers.

[7] The review paper provides a comprehensive analysis of sensor technologies, IoT integration, and machine learning approaches used in hydroponics. It highlights advancements in monitoring systems and their impact on crop yield and resource efficiency. However, the study is largely theoretical and does not provide a practical implementation or real-time validation of the discussed systems.

[8] The paper focuses on monitoring essential environmental parameters using IoT sensors. The system ensures that pH, temperature, and humidity remain within optimal ranges for plant growth. While it improves environmental monitoring, it considers only limited parameters and does not include nutrient concentration monitoring or automated control mechanisms, reducing its overall effectiveness.

[9] The paper presents a smart hydroponic system integrating IoT and machine learning. It monitors multiple parameters and includes image processing techniques for plant health detection. The system enhances decision-making and improves crop productivity. However, it requires high computational resources, increases system cost, and introduces complexity due to image processing and AI integration.

[10] The study presents a hydroponic system with real-time monitoring and mobile-based control. It improves resource utilization and supports sustainable agriculture practices. However, the system primarily focuses on monitoring and lacks complete automation and predictive analysis capabilities.

[11] The paper proposes an advanced system integrating AI and IoT for monitoring nutrient levels and predicting deficiencies. It uses machine learning algorithms to automate nutrient dosing and improve plant health. While the system is highly efficient, it involves high cost, increased complexity, and requires advanced processing capabilities, making it less suitable for basic applications.

[12] The paper introduces an alert-based system that monitors nutrient levels using IoT sensors and provides recommendations for maintaining optimal conditions. It improves nutrient management and reduces manual effort. However, it does not implement full automation and lacks a feedback-based control system for real-time correction.

[13] The review discusses the role of IoT and automation in vertical farming. It highlights the potential of smart farming technologies to improve food security and resource efficiency in urban environments. However, it identifies challenges such as high energy consumption, economic feasibility, and lack of large-scale implementation models.

[14] The paper presents a smart vertical farming system controlled through IoT and mobile applications. It demonstrates improved crop yield and productivity using controlled environmental parameters. However, the system focuses on specific crops, lacks comprehensive sensor integration, and does not include advanced automation or scalability features.

[I] The paper presents an advanced deep learning-based framework for detecting nutrient deficiencies in hydroponic lettuce. The study focuses on improving classification accuracy using an optimized ensemble model that combines multiple dilated networks. This approach enhances the detection of nutrient imbalances and supports precision agriculture. However, the system involves high computational complexity and requires significant processing resources, making it less suitable for low-cost or real-time embedded applications.

[II] The research work introduces an IoT-based nutrient monitoring system capable of measuring specific ion concentrations such as nitrate, ammonium, calcium, and potassium in real time. The system provides detailed insights into nutrient absorption and plant physiological responses. As described on page 1, it uses solid-contact ion-selective electrodes integrated with IoT for continuous monitoring of nutrient solutions. However, the system requires specialized sensors and calibration techniques, which increase cost and complexity for practical deployment.

[III] The paper focuses on improving communication and interoperability among heterogeneous agricultural systems using Industrial IoT (IIoT). The proposed model utilizes Data Distribution Service (DDS) middleware to enable efficient real-time communication between different devices and platforms. The system demonstrates improved throughput and reliable data transmission, as highlighted in the experimental results. However, the implementation is complex and requires advanced networking infrastructure, which may not be feasible for small-scale farming applications.

[IV] The study presents an image processing-based approach for monitoring plant growth using ARUCO markers. The system enables accurate leaf detection, position estimation, and growth analysis through computer vision techniques. It also integrates hydroponic methods such as NFT and DWC for controlled cultivation. While the approach enhances plant monitoring accuracy, it introduces additional hardware requirements such as cameras and increases computational complexity, making it less suitable for cost-sensitive implementations.

[V] The paper provides a comprehensive overview of IoT-enabled smart farming systems in soilless agriculture. It highlights the role of real-time monitoring, automation, and data-driven decision-making in improving crop yield and resource efficiency. As discussed in the introduction, IoT-based systems can significantly reduce water usage and enable controlled environment agriculture. However, challenges such as high initial investment, energy consumption, and technological dependency remain key barriers to widespread adoption.

[VI] The research article presents an AI-integrated hydroponic system designed for cultivating medicinal plants such as holy basil. The system utilizes IoT sensors, cloud computing, and machine learning algorithms to monitor plant health and predict nutrient requirements. It also introduces a “Healthiness Score” for evaluating plant conditions. While the system improves automation and

efficiency, it relies heavily on cloud infrastructure and advanced technologies, which increases system cost and complexity for practical deployment.

3.3 CONCLUSION OF LITERATURE SURVEY

From the literature review performed for this study, it was found that there has been considerable progress made in the area of hydroponic farming systems with IoT applications and automated systems. Many different research journals have been analyzed to determine the current status of monitoring systems employed in hydroponic farms.

According to the findings of the literature review, the majority of systems designed today include the use of IoT-based devices and platforms for real-time monitoring of various parameters like pH levels, humidity, nutrient content, temperature, and water content. Moreover, several research journals also highlighted the implementation of machine learning algorithms and image processing for improving yield and automation in decision-making processes. Although there have been considerable developments in hydroponic systems in terms of monitoring, there are many limitations in terms of automation and closed-loop systems.

From the analysis of previously published research studies, it was found that there is a need for developing a cost-effective and efficient hydroponic farming system that can monitor all parameters in real-time and provide automation capabilities. Hence, for the proposed hydroponic farm system, the ESP32 microcontroller and IoT will be utilized.

CHAPTER 4

METHODOLOGY

4.1 SYSTEM DESIGN AND PLANNING

The system design and planning phase focuses on a structured approach for implementing the IoT-based hydroponic farming and the primary objective is to design system that is capable of monitoring and controlling essential plant growth parameters in real time.

Initially , the critical parameters such as pH level, nutrient concentration, temperature and humidity, and water level are identified. The parameters play a significant role in determining the health of the plant and its productivity in hydroponic systems. Based on standard hydroponic practices, optimal ranges for each parameter are defined.

Appropriate sensors are selected to measure the parameters accurately. The ESP32 microcontroller is chosen as the central control unit due to its built-in Wi-Fi capability, fast processing speed, and suitability for IoT applications. Additional components such as relay modules, water pumps, air pumps, and OLED display were selected to enable automation and real-time monitoring.

A system architecture is designed consisting of sensing, processing control, and communication layers. The system is planned as a closed loop control system, where sensor data is continuously monitored and corrective actions are automatically taken whenever parameters deviate from predefined limits. To design also emphasizes cost-effectiveness, modularity, and scalability so that the system can be easily expanded or modified for different applications.

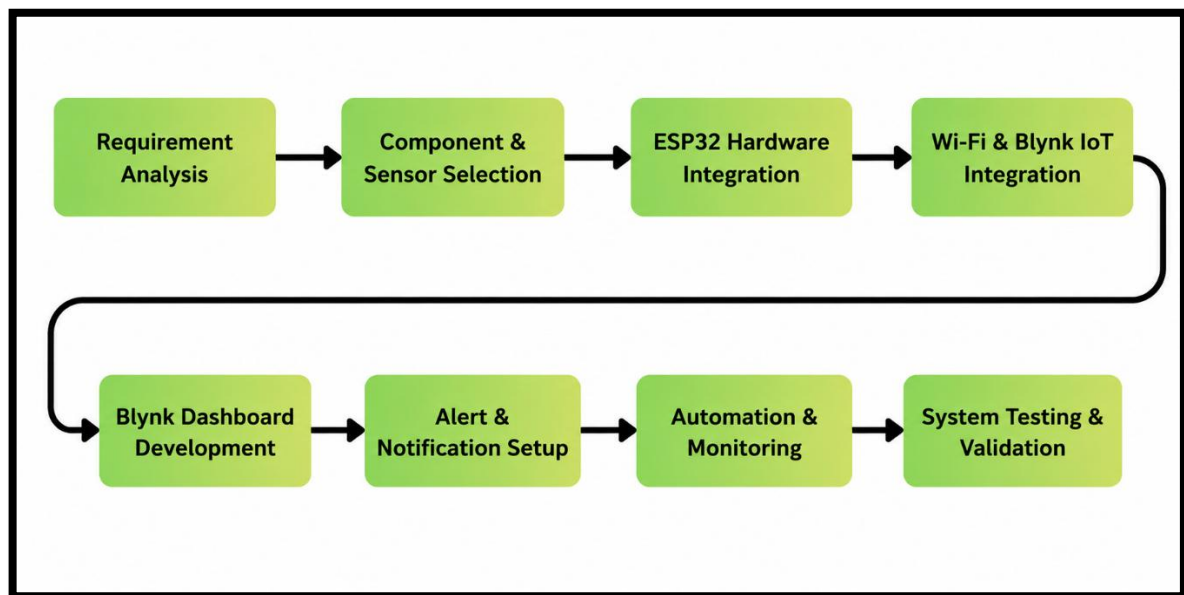


Fig 4.1: Methodology Flow of Proposed System

4.2 HARDWARE IMPLEMENTATION

The hardware implementation involves assembling and interfacing all physical components that are required for the hydroponic system. The ESP32 microcontroller acts as the central unit , and we integrate all the sensors and actuators to the microcontroller.

The following sensors are used:

- pH Sensor: Measures acidity or alkalinity of the nutrient solution
- TDS Sensor: Measures nutrient concentration in water
- DHT22 Sensor: Measures temperature and humidity
- Water Level Sensor: Detects the water level in the reservoir

All the sensors are connected to the ESP32 using appropriate analog and digital pins and the sensor data is continuously read and processed by the microcontroller. The actuation system consists of relay modules connected to the devices like air and water pumps. The relay module acts as an interface that allows ESP32 to control the high-power devices safely. When the parameter deviates from the optimal ranges, the ESP32 sends signals to the relay module to activate or deactivate the respective actuator.

We integrate an OLED display into the system to provide the real-time local visualization of sensor readings. This helps in monitoring system performance without solely relying on the mobile application. All the components are connected properly using a regulated power supply ensuring stable and safe operation. The hardware setup is organized systematically to facilitate easy maintenance and troubleshooting.

4.2.1 ESP32 Microcontroller

ESP32 microcontroller is the core component of the suggested hydroponic farming system. The device reads data from sensors, processing of the collected information, controlling actuators and enabling wireless transmission of the information via Wi-Fi. It is used in IoT solutions due to its powerful processing capabilities and convenient interfacing possibilities.

The proposed hydroponic system will receive the information about pH, TDS, temperature, humidity and water level from sensors all the time. The microcontroller controls various devices like water pump and air pumps based on predetermined parameters. The device will transmit collected data to the Blynk IoT app in real-time mode.

Features

- Built-in Wi-Fi and Bluetooth support
- Dual-core high-speed processor
- Supports multiple communication protocols
- Large number of GPIO pins
- Suitable for IoT and automation applications

Table 4.2.1: Specifications of ESP32 Microcontroller

Sr. No.	Parameter	Specification
1.	Operating Voltage	3.3 V
2.	Wi-Fi Support	802.11 b/g/n
3.	Bluetooth	BLE & Classic Bluetooth
4.	Clock Speed	Up to 240 MHz
5.	GPIO Pins	30+
6.	Communication Protocols	UART, SPI, I2C



Fig 4.2 : ESP32 Microcontroller

4.2.2 pH Sensor

The pH stands for Potential of Hydrogen. The pH sensor is employed to determine the degree of acidity or alkalinity of the nutrient solution used in the hydroponics system. An appropriate pH value is significant since it influences the rate of nutrients absorbed by the plant roots. Poor pH values could lead to slow growth and inefficiency in nutrient intake by plants. The pH sensor constantly monitors the pH value of the nutrient solution and transmits analog values to the ESP32 microcontroller.

Features

- Accurate pH measurement
- Fast response time
- Suitable for water quality monitoring
- Easy interfacing with ESP32
- Continuous real-time monitoring

Table 4.2.2: Specifications of pH Sensor

Sr. No.	Parameter	Specification
1.	Operating Voltage	5V
2.	Measurement Range	0 – 14 pH
3.	Output Type	Analog
4.	Accuracy	±0.1 pH
5.	Response Time	Fast
6.	Application	Water Quality Monitoring



Fig 4.3 pH Sensor

4.2.3 TDS Sensor

TDS sensor helps to measure the quantity of nutrients that are dissolved in the nutrient solution. This is important in order to maintain a proper nutrient concentration in the nutrient solution. The TDS sensor keeps track of the concentration of nutrients and provides an analog output value to the ESP32 microcontroller. These values are helpful in determining whether any adjustments are necessary regarding the nutrient concentration.

Features

- Measures nutrient concentration in water
- Real-time monitoring capability
- Analog output support
- Easy integration with ESP32
- Suitable for hydroponic applications

Table 4.2.3: Specifications of TDS Sensor

Sr. No.	Parameter	Specification
1.	Operating Voltage	3.3V- 5V
2.	Measurement Range	0 – 1000 ppm
3.	Output Type	Analog
4.	Compatibility	ESP32/Arduino
5.	Response Time	Fast
6.	Application	Water Quality Monitoring

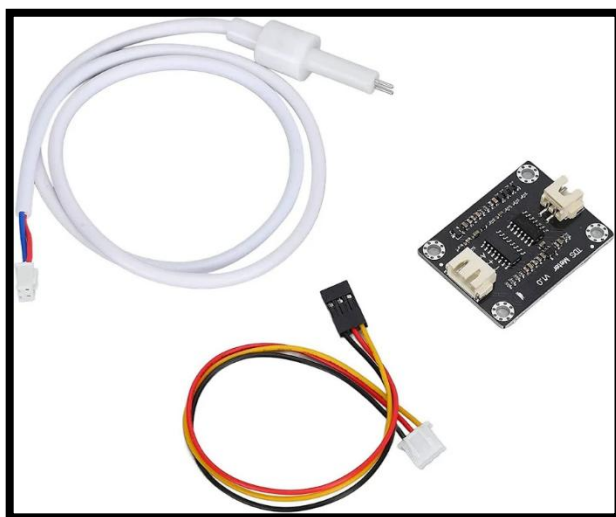


Fig 4.4 TDS Sensor

4.2.4 DHT22 Sensor

The DHT22 sensor is employed to monitor the environmental parameters including temperature and humidity in the hydroponics system. Proper management of environmental parameters like temperature and humidity is required for plant growth and higher crop production.

The readings provided by the DHT22 sensor are accurate and stable. Data acquired from environmental sensors is analyzed using the ESP32 board and is shown both locally on the OLED display and remotely on the Blynk app.

Features

- Measures both temperature and humidity
- High measurement accuracy
- Digital signal output
- Stable and reliable performance
- Low power consumption

Table 4.2.4: Specifications of TDS Sensor

Sr. No.	Parameter	Specification
1.	Operating Voltage	3.3V- 5V
2.	Temperature Range	-40°C to 80°C
3.	Humidity Range	0 – 100% RH
4.	Output Type	Digital
5.	Compatibility	ESP32/Arduino
6.	Accuracy	±0.5°C

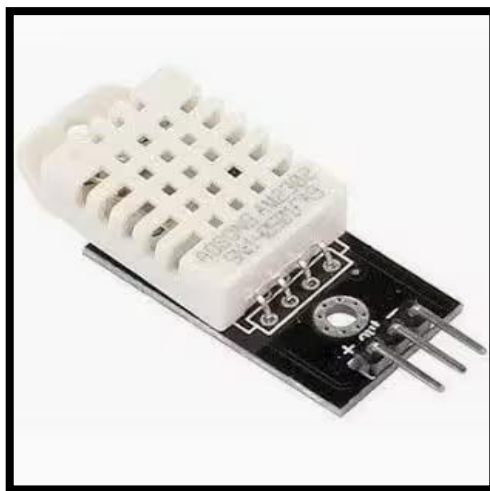


Fig 4.5 DHT22 Sensor

4.2.5 Water Level Sensor

The water level sensor detects the amount of nutrient solution contained within the reservoir. It is important to monitor the water level to ensure that the nutrients continue to circulate and avoid drying up the pump. The sensor continually checks the water level and sends signals to the microcontroller ESP32. When the water level falls below the required threshold, it turns on the pump or triggers an alarm.

Features

- Real-time water level detection
- Fast response time
- Simple interfacing with ESP32
- Suitable for liquid monitoring
- Reliable operation

Table 4.2.5: Specifications of Water Level Sensor

Sr. No.	Parameter	Specification
1.	Operating Voltage	3.3V- 5V
2.	Response Time	Fast
3.	Sensitivity	Adjustable
4.	Output Type	Analog/Digital
5.	Compatibility	ESP32/Arduino
6.	Application	Water Level Detection

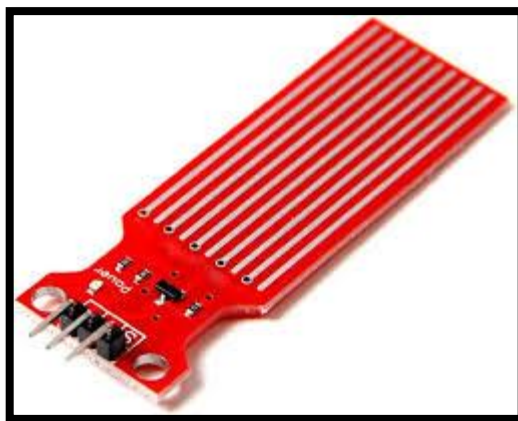


Fig 4.6 Water Level Sensor

4.2.6 Relay Module

A single-channel relay module is used to control the air pump in the hydroponic farming system. The relay acts as an electrically operated switch that allows the ESP32 microcontroller to control higher-power devices safely. It provides electrical isolation between the low-voltage control circuit and the air pump, improving system reliability and safety. The relay receives digital control signals from the ESP32 and switches the air pump ON or OFF whenever required.

Features

- High power switching ability
- Electrical isolation possible
- Digital control easy
- Reliable performance

Table 4.2.6: Specifications of Relay

Sr. No.	Parameter	Specification
1.	Operating Voltage	5V DC
2.	Number of Channels	Single Channel
3.	Isolation	Optocoupler Isolation
4.	Trigger Type	Digital
5.	Application	Pump Control

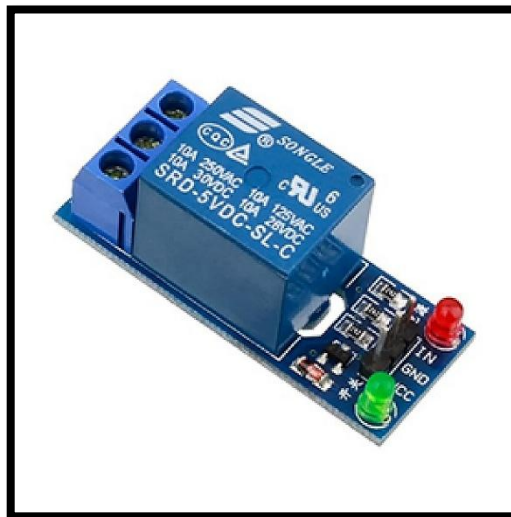


Fig 4.7 Single Channel Relay

4.2.7 Water Pump

Water Pump 1 – Automatic Refill Pump

This pump is used to maintain the required water level in the nutrient reservoir. The water level sensor continuously monitors the reservoir level. Whenever the water level falls below the predefined threshold, the ESP32 activates the refill pump through the relay module. The pump transfers water into the reservoir until the desired level is restored.

Water Pump 2 – Drainage Pump

The second water pump is used for draining or removing nutrient solution from the reservoir during maintenance, cleaning, or nutrient replacement. This pump is operated manually by the user whenever required and is not controlled automatically by the ESP32.

Features

- Continuous nutrient circulation
- Low power consumption
- Compact structure
- Suitable for hydroponic systems

Table 4.2.7: Specifications of Water Pump

Sr. No.	Parameter	Specification
1.	Operating Voltage	6-9V DC
2.	Pump Type	Submersible Pump
3.	Flow Rate	Moderate Flow

4.2.8 Air Pump

The air pump is used to supply oxygen to the nutrient solution. Adequate oxygen levels improve root respiration and support healthy plant growth within the hydroponic system.

Features

- Provides oxygen supply
- Improves root health
- Low power consumption
- Continuous operation support

Table 4.8: Specifications of Air Pump

Sr. No.	Parameter	Specification
1.	Operating Voltage	6-9V DC
2.	Pump Type	Aquarium Air Pump
3.	Function	Oxygen Supply

4.2.9 OLED Display

The OLED screen enables visualization of the sensor readings locally and instantly. The screen shows various parameters of the system, which include PH, TDS, temperature, humidity, and water level on the hardware device.

The OLED screen makes it convenient for users to check the parameter readings instantaneously and without solely relying on the mobile app.

Features

- Compact display size
- Real-time data visualization
- Low power consumption
- High display clarity
- I2C communication support

Table 4.9: Specifications of OLED Display

Sr. No.	Parameter	Specification
1.	Display Type	OLED
2.	Communication	I2C
3.	Display Size	0.96 inch
4.	Resolution	128 x 64 Pixels
5.	Operating Voltage	3.3V to 5V
6.	Application	Real Time Display



Fig 4.8 OLED Display

4.3 SOFTWARE DEVELOPMENT

It should be noted that software development is a crucial part of the realization of the proposed system. It involves the collection of sensor data, processing of information, actuator control, and the establishment of communication between the ESP32 microcontroller and the Blynk IoT server. The system software code is written in the Arduino IDE. The Blynk platform is utilized to provide remote monitoring and control of the system.

ESP32 firmware is written in C. Different libraries are incorporated in the software to interface with sensors, Wi-Fi communication, OLED display control, and IoT functionality. The software reads real-time data from different sensors regarding the pH level, nutrient concentration, temperature, humidity, and water level. The values are processed and compared to the corresponding predefined threshold ranges to see whether the hydroponic system is functioning in favorable conditions. If the sensor values fall outside the thresholds, the ESP32 decides what measures have to be taken and automatically starts control of the actuators through the relay module. For instance, if the water level decreases to a certain extent, the water pump is started automatically. Notifications are sent in case of abnormalities in the environment.

4.3.1 Arduino Integrated Development Environment (IDE)

Arduino IDE is a free-to-use software tool designed for the creation of program code to be uploaded on the ESP32 microcontroller board. Arduino IDE offers users an easy way of writing and uploading code onto the board.

In the proposed system, the Arduino IDE will help design the program to control the sensors, relay switch, Wi-Fi connectivity, management of the OLED screen, and connection with Blynk application. The software will also facilitate serial monitoring that allows for debugging and analysis of values from the sensor. The software offers different libraries that ease the communication process between the ESP32 board and other hardware devices connected to it.

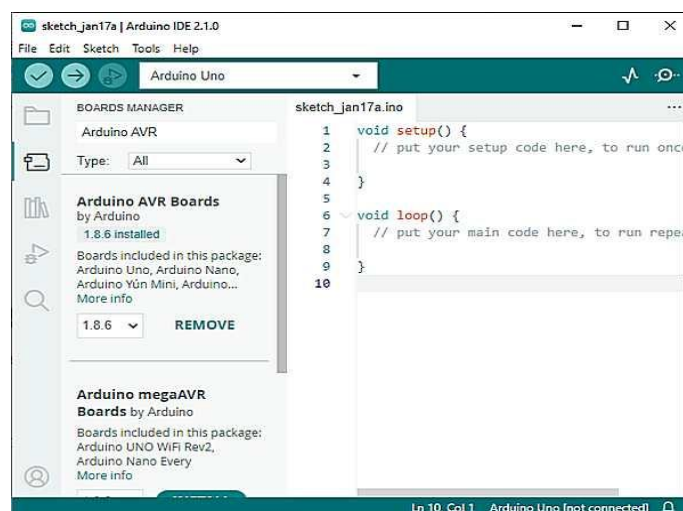


Fig 4.9 Arduino IDE Software

4.3.2 Blynk IoT Platform

Remote monitoring and control of the hydroponic system are performed using Blynk IoT platform via mobile applications. Blynk provides real-time visualization of sensor readings and allows the users to access information about the system from anywhere in the world through the Internet.

The ESP32 module sends readings obtained by sensors like pH, temperature, humidity, nutrients concentration, and water level to Blynk cloud platform. These sensor readings are displayed on the Blynk dashboard via widgets, graphs, gauges, and alerts sections.

Blynk offers additional options like alert generation and remote control that make the system more convenient and accessible for users. Users can constantly monitor the performance of the system by using Blynk mobile application and get alerts if some reading goes above certain values.

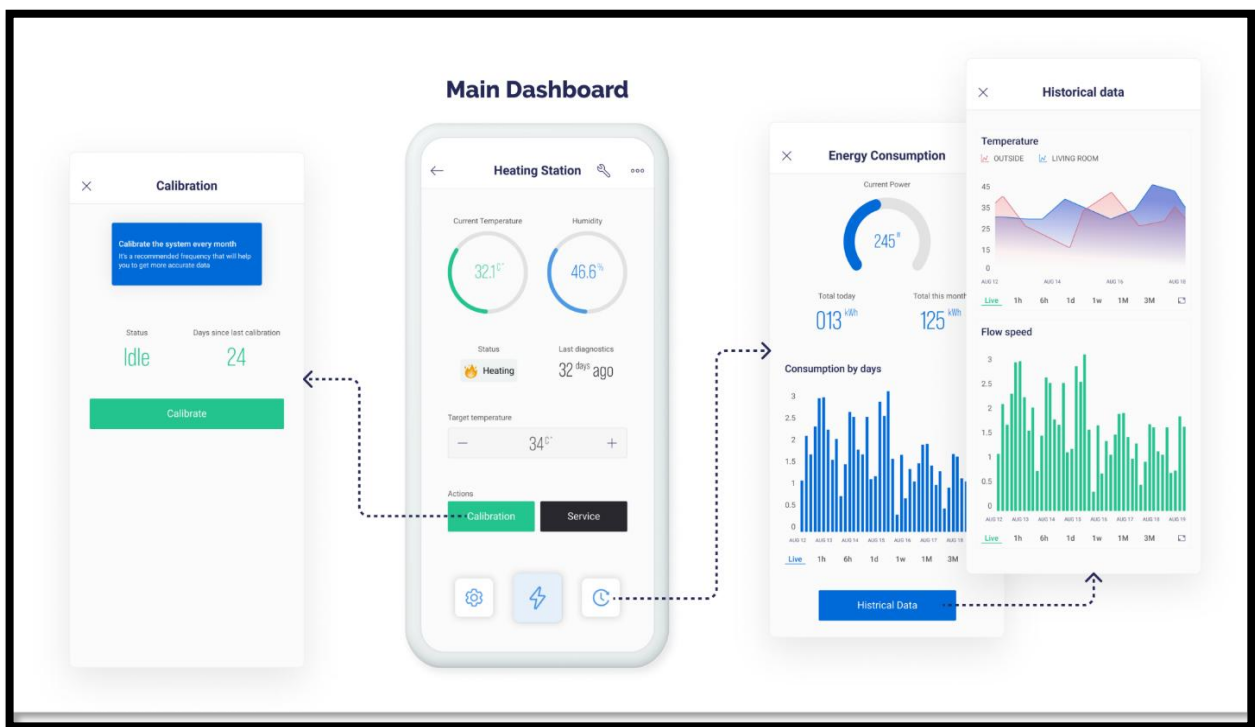


Fig 4.10 Blynk IoT

4.3.3 Proteus Simulation Software

Proteus is an electrical circuit simulation tool that is commonly used for validation of an embedded system prior to any physical implementation. The program offers virtual testing of an electronic circuit and enables the detection of potential errors at the design stage. Within the framework of the project suggested, the hydroponic farming circuit will be simulated using the Proteus software to check the performance of all sensors, relays, and controllers involved. The simulation will help to check the circuit performance prior to its physical implementation.

The advantage of the Proteus program is that there is a possibility to perform virtual tests and evaluate the performance of the system based on the ESP32 microcontroller under different conditions.



Fig 4.11 Proteus Simulation Software

4.4 SYSTEM INTEGRATION, DEPLOYMENT AND MONITORING

In this phase, all hardware and software components are integrated to form a complete working system. The sensors, actuators, ESP32, and communication modules are interconnected and tested collectively to ensure proper coordination. The system is deployed in a hydroponic setup consisting of a nutrient reservoir, grow tray, and circulation system. Once deployed, the system continuously monitors environmental and nutrient parameters. Sensor readings are analyzed in real time, and necessary actions are taken automatically to maintain optimal conditions.

The system provides dual monitoring capability:

- Local monitoring through OLED display
- Remote monitoring through Blynk mobile application

Users can observe system performance, receive alerts, and control actuators remotely. The system is tested under different conditions to evaluate its response, accuracy, and stability. Performance is analyzed by comparing measured values with expected ranges and observing plant growth behavior. Calibration is performed when necessary to improve accuracy. The integrated system successfully demonstrates efficient automation, reduced manual intervention, and improved resource management, making it suitable for practical hydroponic farming applications.

The system incorporates a real-time notification mechanism to enhance monitoring capabilities. Whenever the water level sensor detects that the nutrient reservoir level falls below the predefined threshold, the ESP32 microcontroller triggers an alert through the Blynk IoT platform. Simultaneously, an email notification is sent to the user, informing them about the low water level condition. This feature ensures timely corrective action and prevents interruptions in plant growth due to insufficient nutrient solution.

CHAPTER 5

IMPLEMENTATION

5.1 SYSTEM OVERVIEW

The implementation of the proposed system involves the practical realization of an IoT-based hydroponic farming setup using the ESP32 microcontroller. The system integrates multiple sensors, actuators, wireless communication, and a cloud-based platform to monitor and maintain environmental and nutrient conditions required for healthy plant growth. The primary objective of the implementation is to develop a reliable system capable of real-time monitoring, automation, and remote accessibility.

The system continuously monitors important parameters such as pH level, Total Dissolved Solids (TDS), temperature, humidity, and water level using dedicated sensors. The collected data is processed by the ESP32 microcontroller and displayed locally on an OLED display while simultaneously being transmitted to the Blynk IoT platform through Wi-Fi connectivity for remote monitoring.

The implementation utilizes two water pumps for water management. The first pump is used to refill the nutrient reservoir whenever a low water level condition is detected, while the second pump is operated manually for draining water during maintenance and cleaning operations. An air pump is also incorporated to provide adequate oxygen to the nutrient solution and is controlled through a single-channel relay module interfaced with the ESP32.

To improve system reliability, a notification mechanism has been integrated using the Blynk IoT platform. Whenever the water level falls below the predefined threshold, the system generates a push notification on the Blynk application and simultaneously sends an email alert to the user. The implemented system forms an intelligent hydroponic farming setup capable of maintaining suitable growing conditions with minimal human intervention while providing remote monitoring and timely alerts.

5.2 BLOCK DIAGRAM

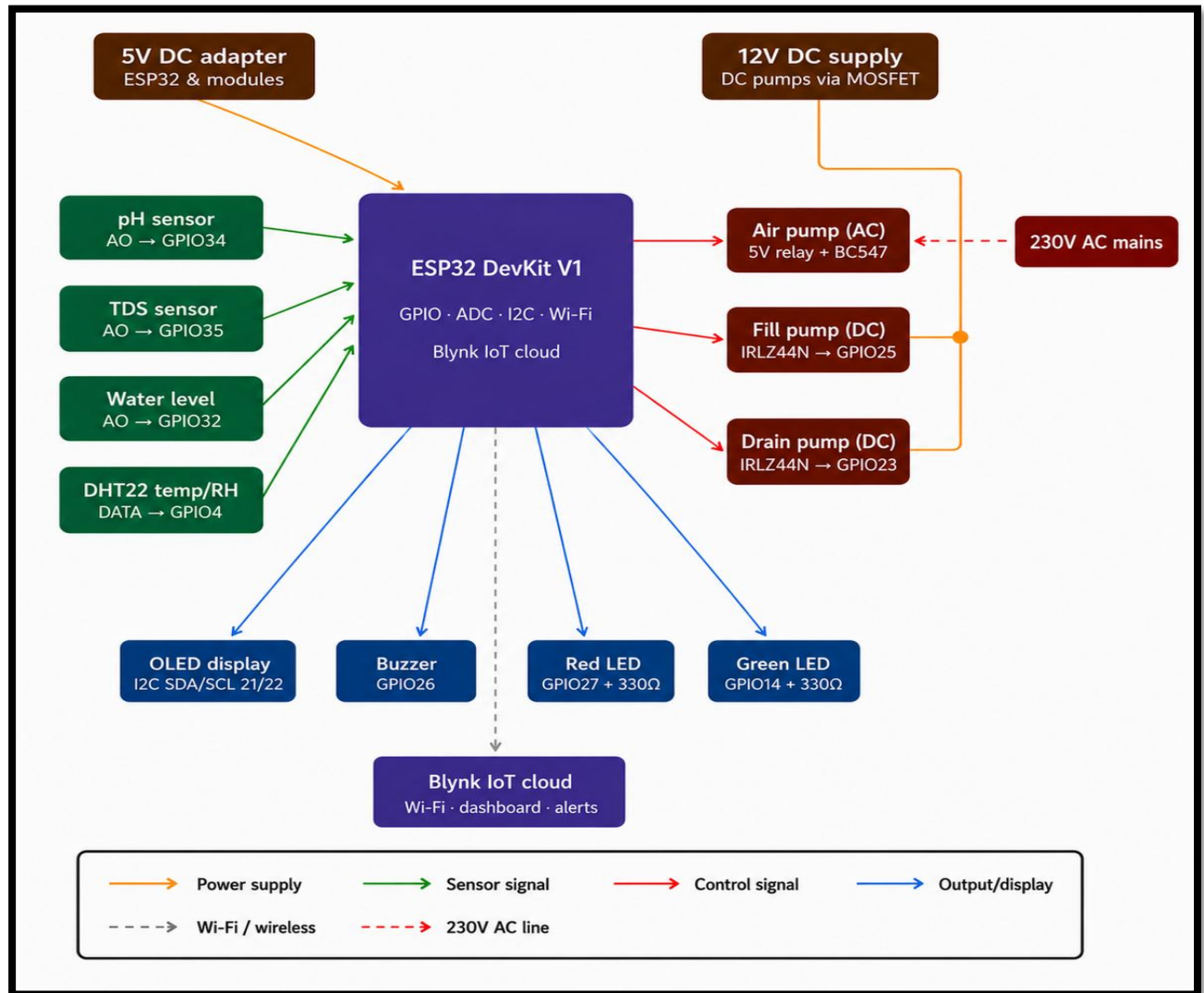


Fig 5.1 Block Diagram of Hydroponics System

The block diagram for the proposed hydroponic farm shows the structure and interaction among the different components that make up the system. The system basically comprises sensors, ESP32 microcontroller, actuators, user interface modules, power supply, and the hydroponic growing tank.

Sensors such as pH sensor, TDS sensor, DHT22 sensor, and water level sensor monitor different aspects of the environment and nutrient composition. Data from the sensors is fed into the central microcontroller in the form of the ESP32. The sensor data is analyzed by the ESP32 against the defined threshold.

According to the data obtained from analysis, the central controller triggers the activation or deactivation of both the water pump and air pumps using relay. Sensor readings are visualized using the OLED display while remote monitoring is enabled via Blynk IoT application through Wi-Fi communications. The ESP32 microcontroller communicates with the Blynk IoT application through its built-in Wi-Fi module, enabling real-time monitoring of sensor data and transmission of low water level notifications and email alerts to the user.

5.3 CIRCUIT DIAGRAM

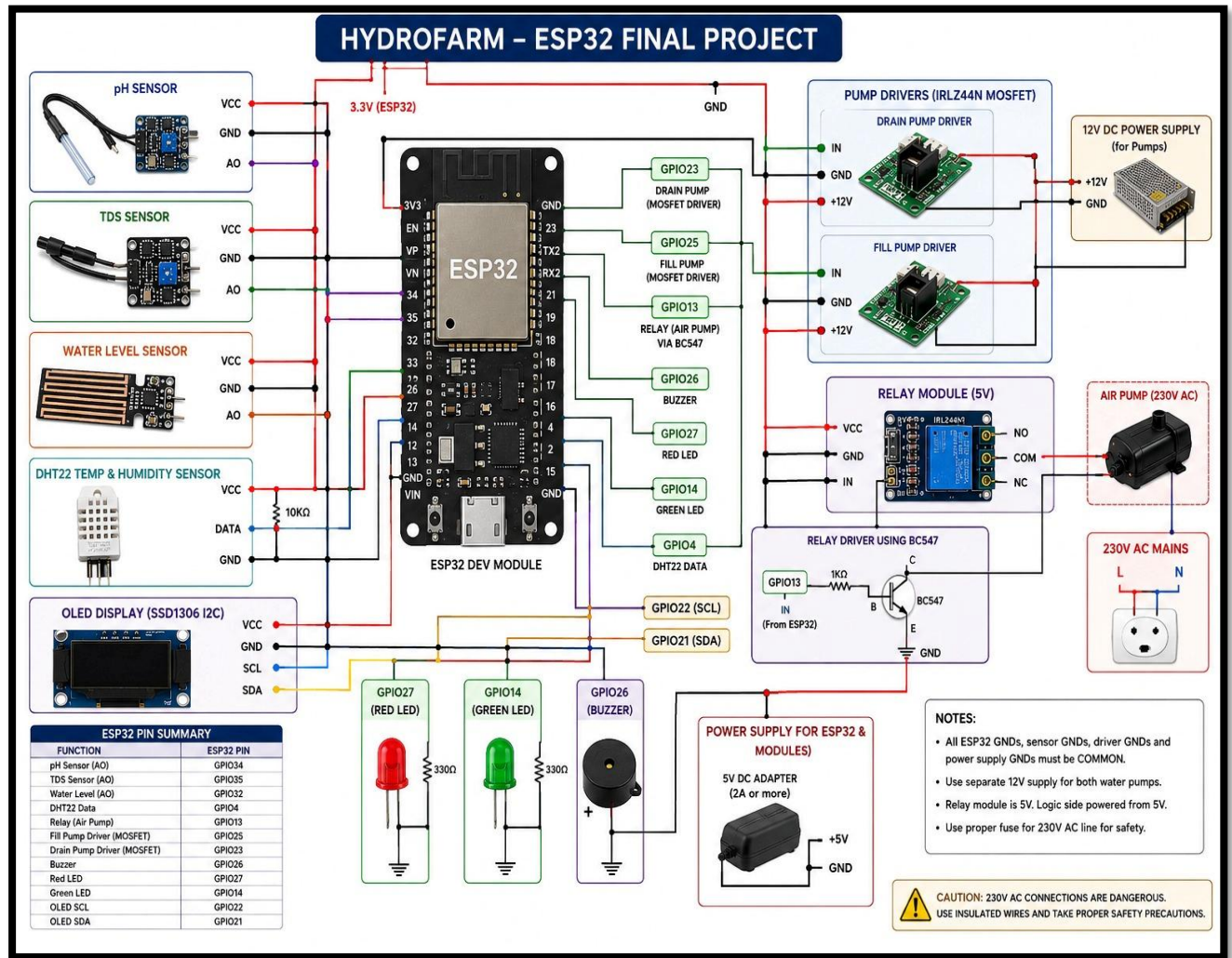


Fig 5.2: Circuit Diagram of Proposed System

The circuit diagram below represents the electric connection of sensors, relay modules, OLED display, and actuators with the ESP32 microcontroller. All components are electrically connected using the correct GPIO pins and power supply connections for efficient functioning of the whole system.

The pH sensor, TDS sensor, DHT22 sensor, and water level sensor are connected to the input pins of the ESP32 for the continuous monitoring of the environment and nutrients. Relay module connects to the ESP32 in order to control water pump and air pump depending on the results obtained from the sensors.

OLED display connects through the I2C protocol so as to display real-time sensor data on the hardware setup. ESP32 also connects to Blynk platform wirelessly by using Wi-Fi for remote monitoring.

5.4 FLOWCHART

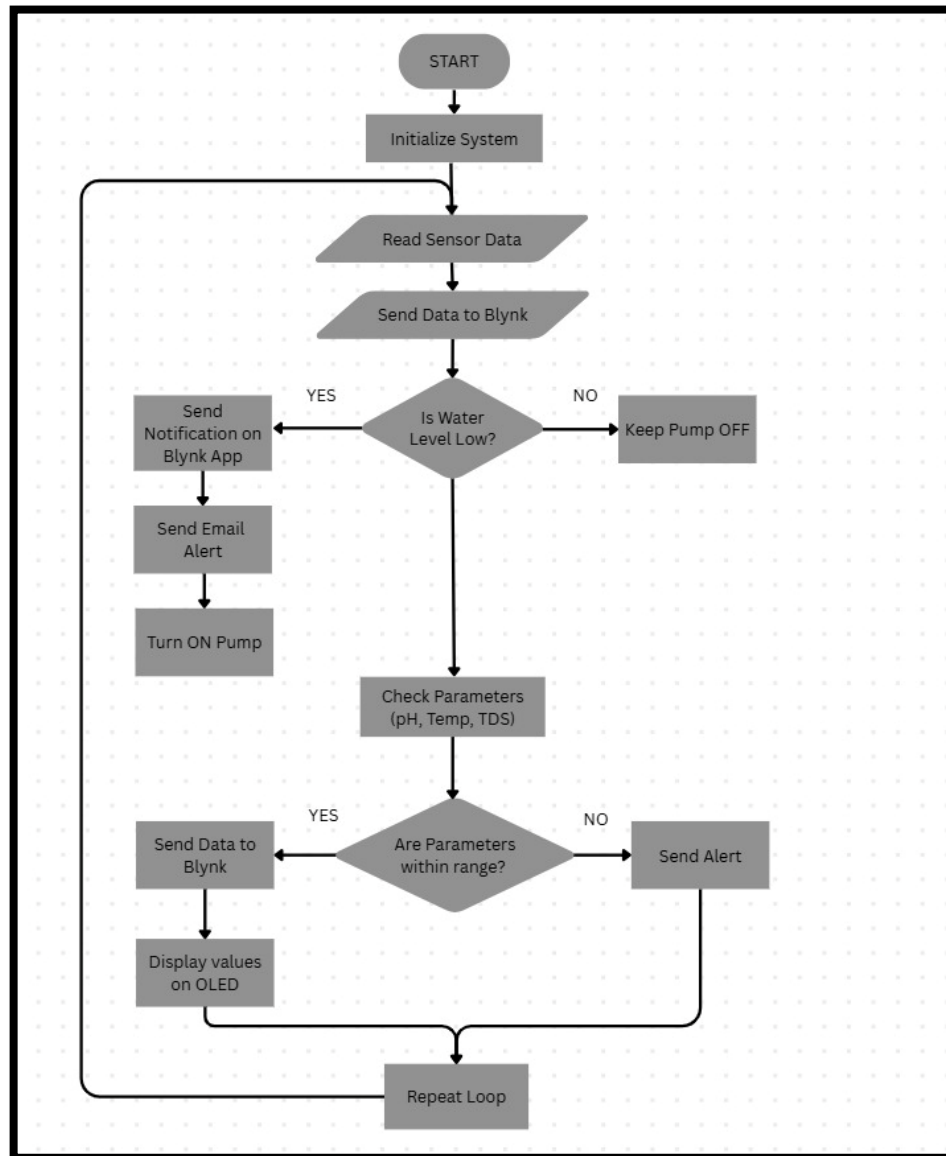


Fig 5.3: Flowchart of System Operation

The flowchart shows the operational flow of the designed hydroponic farming technique. This flowchart describes the working mechanism of the proposed monitoring and automation technique carried out by the ESP32 microcontroller. At first, the whole system gets started, and all kinds of sensors, communication devices, and displays get initialized. In this way, sensors start measuring the real-time data corresponding to pH level, nutrient levels, temperature, humidity, and water level in the hydroponic system.

The ESP32 processes the real-time sensor measurements and compares them with the pre-programmed ranges suitable for plant growth. When any sensor measurement lies outside the suitable range, then the ESP32 automatically initiates the control action. For instance, when the water level decreases below the threshold value, the ESP32 microcontroller automatically turns on the water pump. The processed sensor data is shown on the OLED display screen and also sent to the Blynk IoT app.

5.5 HARDWARE SETUP

The hardware configuration for the proposed hydroponic farming system includes the use of an ESP32 microcontroller, pH meter, TDS meter, DHT22 meter, water level meter, relay board, OLED display, water pump, air pump, and hydroponics tank setup. All hardware devices are wired together appropriately and supplied by regulated power supplies. All sensors are carefully placed inside the hydroponics tank for precise monitoring.

The relay board connects between the ESP32 board and both pumps to allow automated switching functions. The OLED display is attached to the hydroponics setup for onsite monitoring, whereas Wi-Fi communication facilitates offsite monitoring via the Blynk IoT system. The entire setup is expected to run continuously under live operational conditions to provide optimal growth conditions for hydroponics farming.



Fig. 5.4: Hardware Setup

5.6 WORKING OF SYSTEM

The working of the proposed Hydroponic Farming System is based on real-time monitoring, automation, and IoT-based communication. The entire system is controlled by the ESP32 microcontroller, which acts as the central processing unit. Initially, the ESP32 initializes all connected sensors, communication modules, and the OLED display. The sensors continuously monitor important parameters such as pH level, nutrient concentration (TDS), temperature, humidity, and water level within the hydroponic setup.

The sensor readings are collected and processed by the ESP32 microcontroller. The measured values are compared with predefined threshold values suitable for hydroponic cultivation. Based on this comparison, the ESP32 performs automatic control actions. When the water level falls below the specified threshold, the water pump is activated to maintain the required nutrient solution level. The system utilizes two water pumps for efficient water management. The first pump operates automatically and is activated when the water level sensor detects a low water condition in the reservoir. The second pump is used manually to drain water from the hydroponic system during maintenance activities such as cleaning, nutrient replacement, or reservoir servicing. Similarly, the air pump is operated to ensure adequate oxygen supply to the plant roots.

The monitored data is displayed locally on the OLED display and simultaneously transmitted to the Blynk IoT platform through Wi-Fi communication. Users can remotely monitor all system parameters using the Blynk mobile application.

To enhance system reliability, a notification and alert mechanism is incorporated. Whenever the water level drops below the predefined threshold, the ESP32 triggers a push notification on the Blynk application and simultaneously sends an email alert to the user. These alerts enable timely corrective action and prevent interruptions in plant growth due to insufficient nutrient solution. The system operates continuously in a loop, ensuring efficient monitoring, timely notifications, and maintenance of optimal environmental conditions for sustainable hydroponic farming.

CHAPTER 6

RESULT AND ANALYSIS

6.1 SIMULATION RESULTS

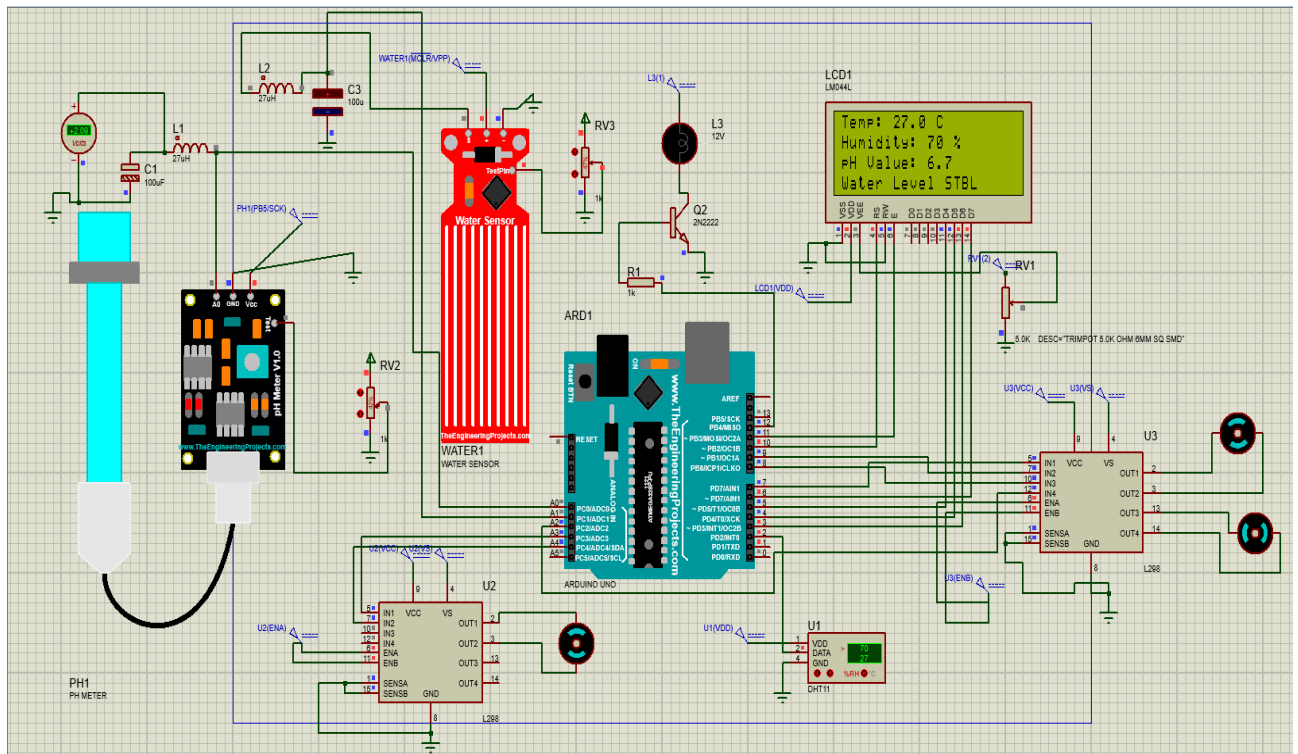


Fig 6.1: Proteus Simulation of Proposed System

The designed hydroponics farming system was modelled using Proteus before the hardware implementation. The modelling of the system was done to validate the interface between the sensors, relays, OLED screen, and the ESP32.

From the simulation, proper interfacing was achieved and verified in terms of the ESP32 controlling its peripherals. The relay module worked perfectly based on the state of the sensors while other parts of the program executed perfectly. This enabled identification of any problems during the hardware process. The results from the simulation showed the successful operation of the hydroponics farming automation system.

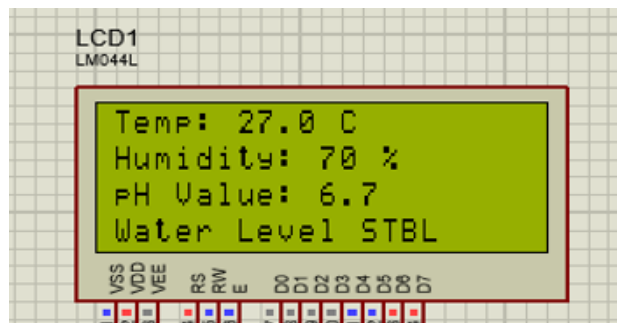


Fig 6.2 LCD output during Simulation

6.2 EXPERIMENTAL SETUP

The experimental setup of the proposed system consists of an ESP32 microcontroller, pH sensor, TDS sensor, DHT22 sensor, water level sensor, relay module, OLED display, water pump, air pump, and hydroponic reservoir setup.

The sensors were placed strategically within the hydroponic setup to continuously monitor environmental and nutrient conditions. The ESP32 microcontroller collected and processed the sensor readings and controlled the connected actuators automatically through relay modules.

The Blynk IoT platform was integrated with the system to provide real-time remote monitoring through a mobile application. The OLED display provided local visualization of sensor values directly on the setup. The complete system was tested continuously to evaluate monitoring accuracy, automation response, and communication reliability.



Fig 6.3: Experimental Hydroponic Setup



Fig 6.4: Sensor Integration Setup

6.3 OBSERVATION TABLE

Table 6.1: Ideal Parameter Ranges

Parameter	Ideal Values
pH Level	5.5-6.8
Temperature	23- 35°C
Humidity	50 – 70 %
TDS Value	400 – 900 ppm
Water Level	Normal

Table 6.2: Observed values

Day	pH	TDS (ppm)	Temperature (°C)	Humidity (%)	Water Level	Remarks
1	6.2	710	27	63	Normal	Normal
2	6.3	715	28	64	Normal	Normal
3	6.4	720	30	65	Normal	Normal
4	6.3	725	34	62	Normal	High Ambient Temp
5	6.4	718	35	60	Low	Alert Generated
6	6.2	722	33	61	Normal	Stable
7	6.3	730	29	66	Normal	Normal
8	6.4	725	28	67	Normal	Normal
9	6.3	720	27	65	Normal	Normal
10	6.4	728	28	66	Normal	Normal

The observation table shows the real-time sensor readings obtained during system testing. The collected values remained within acceptable ranges suitable for hydroponic cultivation, demonstrating stable operation of the proposed system.

6.4 GRAPHICAL ANALYSIS

The collected sensor readings were analyzed graphically to observe the variation of environmental and nutrient parameters during system operation. The graphical analysis helped evaluate the stability and consistency of the monitoring system.

The obtained graphs showed that the proposed system was capable of maintaining stable pH levels, nutrient concentration, temperature, and humidity within the desired range. The automation mechanism responded effectively whenever parameter variations occurred.

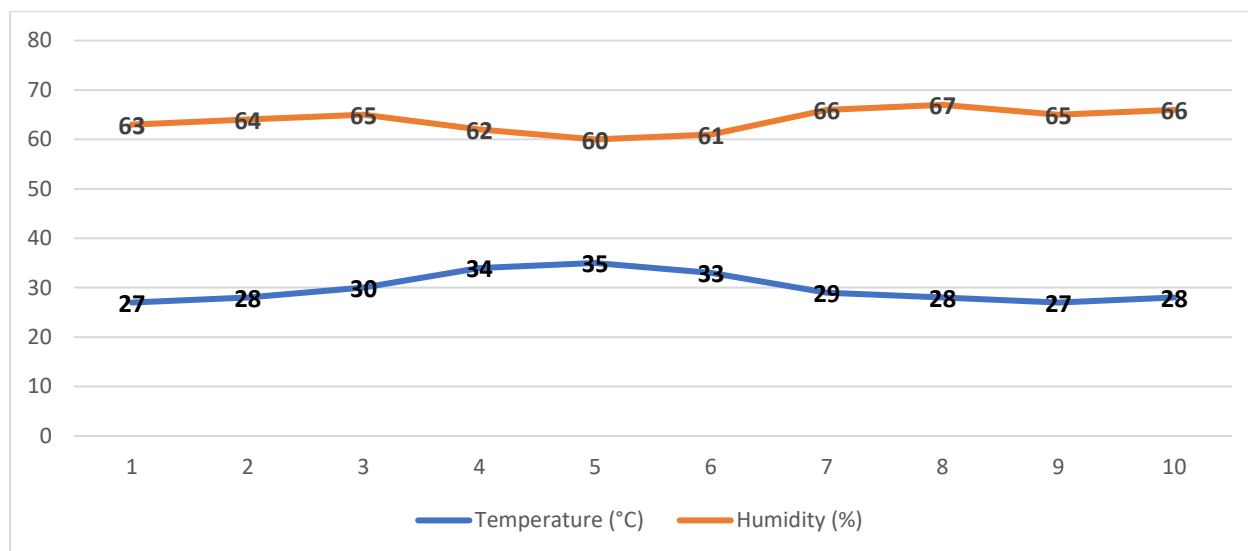


Fig 6.5: Graph of Temperature and Humidity variation

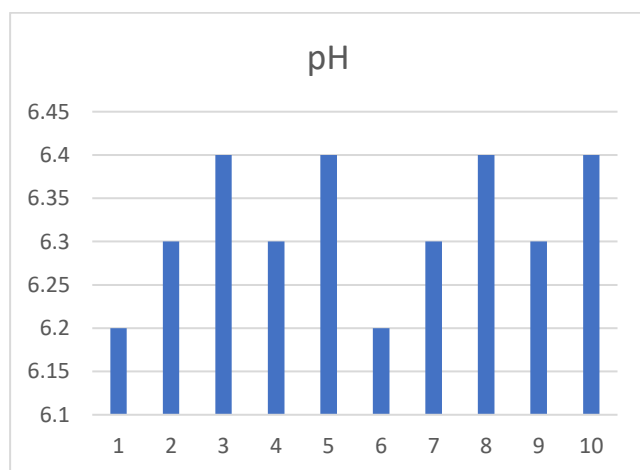


Fig 6.6: Graph of observed pH range

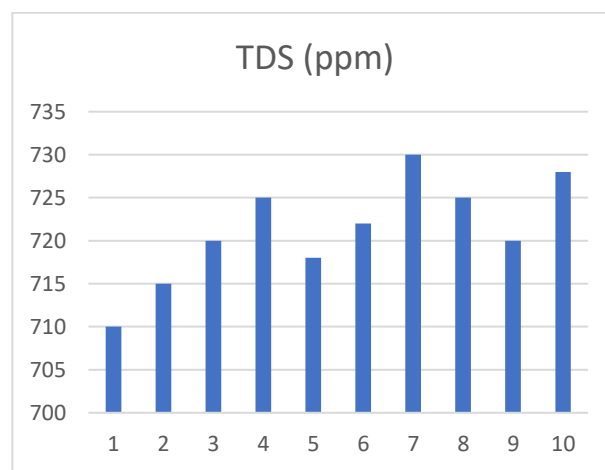


Fig 6.7: Graph of observed TDS range

6.5 BLYNK IOT MONITORING RESULTS

The Blynk IoT platform successfully displayed real-time sensor readings through the mobile application. The dashboard provided graphical visualization of pH level, nutrient concentration, temperature, humidity, and water level data.

The application also supported alert notifications whenever abnormal conditions were detected within the hydroponic setup. Remote accessibility through Wi-Fi communication improved monitoring convenience and reduced the need for continuous manual observation.

The real-time monitoring capability of the Blynk platform improved overall system efficiency and enabled effective management of the hydroponic farming environment.

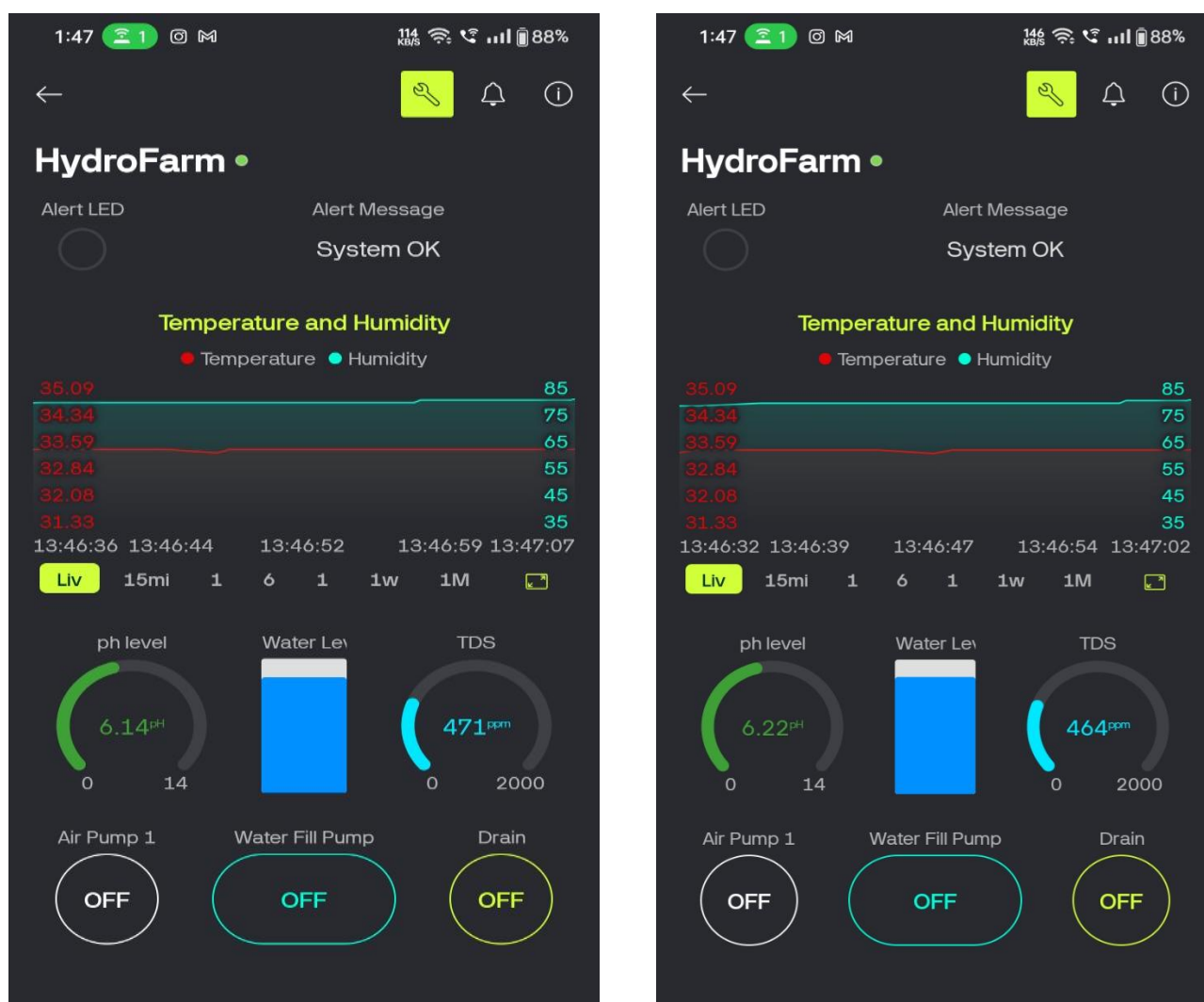


Fig 6.8: Blynk IoT Dashboard

6.5 NOTIFICATION AND ALERT SYSTEM

The notification system was tested by intentionally reducing the water level in the nutrient reservoir. When the water level dropped below the predefined threshold, the ESP32 successfully triggered a push notification on the Blynk mobile application and simultaneously generated an email alert. The notifications were received within a few seconds, demonstrating the reliability of the remote monitoring feature. This functionality ensures that users are immediately informed about critical conditions even when they are away from the hydroponic setup.

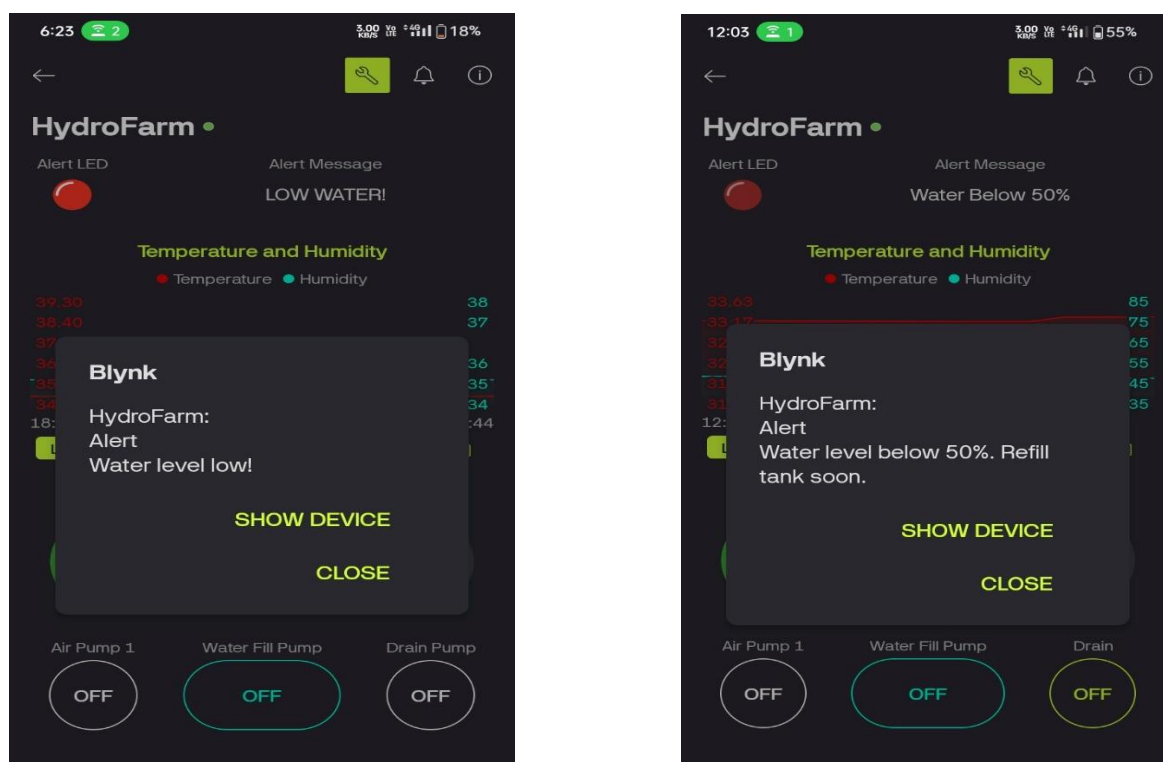


Fig 6.9: Blynk notification screenshot

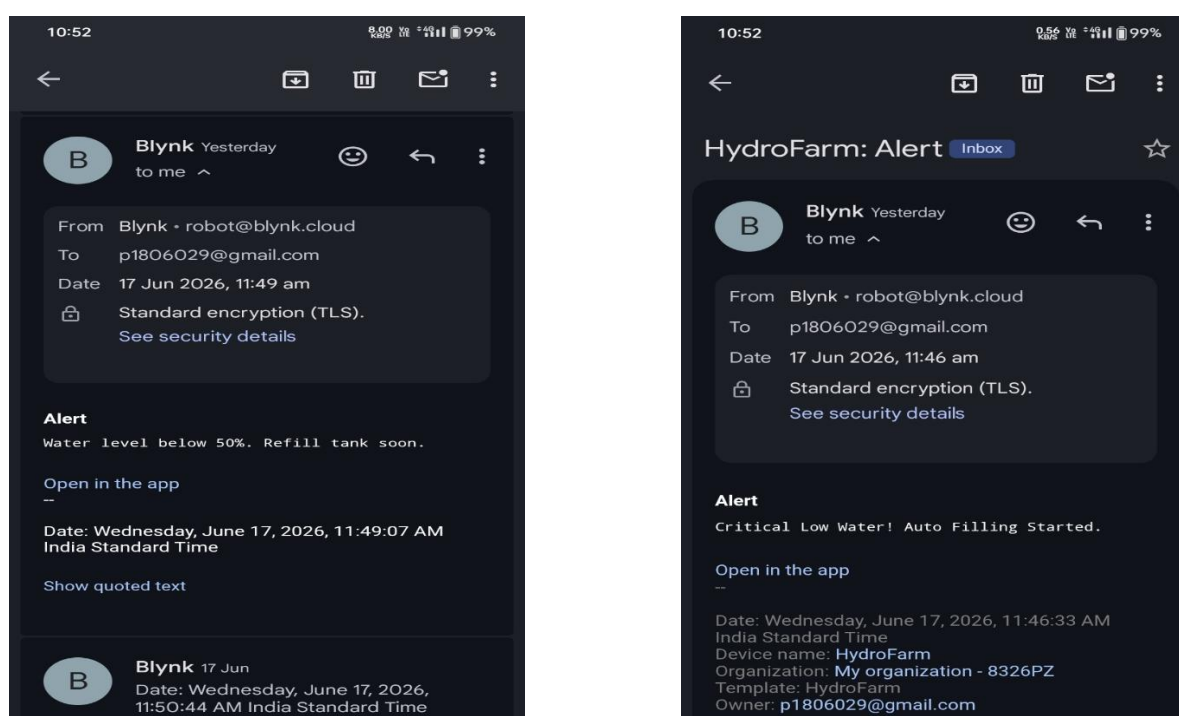


Fig 6.10: Email alert screenshot



CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The design, development, testing, and implementation of the intelligent hydroponic farming system using ESP32 were successful. The ESP32 microcontroller was able to continuously monitor various environmental and nutrient conditions, such as pH levels, nutrient concentration, temperature, humidity, and water level using several sensors. The sensor data was accurately processed by the ESP32 microcontroller, and it was able to perform various automated controls based on pre-defined thresholds using relay modules and pumps.

The inclusion of the Blynk Internet of Things (IoT) platform facilitated remote monitoring using a mobile app. Furthermore, the use of an OLED display provided an alternative way of visualizing the values from the sensors directly from the hardware setup. The developed hydroponic farming system demonstrated effective automation, minimized human intervention, enhanced water management, and efficient monitoring performance in real-time conditions.

7.2 FUTURE SCOPE

Even though the suggested hydroponics farming system showed great success in operation, there are numerous advancements and improvements that could be implemented in future developments to increase the efficiency of the system and its scalability.

AI and Machine Learning techniques can be added to predict plant growth conditions and manage nutrients, thus improving decision-making processes. The cloud computing technique can be used to collect, store, and analyze data on a larger scale for a better performance analysis in the future. The hydroponics farming system can be improved to accommodate commercial farming operations through the addition of more sensors, automated nutrient dosing, and advanced environmental management techniques. Moreover, solar energy can be utilized to reduce the dependency on regular power sources for the system.

Moreover, future improvements in the system can include mobile notifications, voice assistant capabilities, picture-based plant health detection, disease detection system implementation, and automatic nutrient balancing system implementation. In addition, the wireless network communication technology and cloud-based dashboards can be added to improve access remotely and manage smart agriculture effectively.

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